E-COMMERCE PERFORMANCE ANALYTICS DASHBOARD

Complete Project Documentation

Analyst	Sabarivenkatesh Kathirvel
Dataset	Olist Brazil E-Commerce (2016-2018)
Records	100,000+ real e-commerce orders
Tools	Python | SQL | SQLite | Power BI | GitHub
Generated	April 03, 2026

1. PROJECT OVERVIEW

This project is a complete end-to-end data analytics solution built on real-world e-commerce data. The entire workflow — from raw CSV ingestion to an interactive Power BI dashboard with automated insight generation — was designed to match the exact skill requirements expected by top data analytics companies including Tiger Analytics, Fractal Analytics, Swiggy, Razorpay, LatentView Analytics, and Accenture.

The project demonstrates the full data analyst stack: database design, SQL querying, Python-based data cleaning and EDA, business visualization, and automated reporting — all documented in a professional GitHub repository.

Business Questions Answered

- Which product categories generate the most revenue?
- What is the on-time delivery rate and where does it fail?
- When do customers order most — by day and by hour?
- Which seller cities drive the most sales volume?
- What does customer satisfaction data reveal about the business?
- What are the top strategic recommendations for growth?

2. DATASET

The Olist Brazilian E-Commerce Public Dataset contains 100,000 real orders placed between 2016 and 2018 across multiple Brazilian marketplaces. It consists of 9 relational CSV files covering the complete order lifecycle from purchase to delivery and customer review.

Table Name	Rows	Description
olist_orders_dataset	99,441	Core order information
olist_customers_dataset	99,441	Customer location details
olist_order_items_dataset	112,650	Products within each order
olist_order_payments_dataset	103,886	Payment transactions
olist_products_dataset	32,951	Product catalog
olist_sellers_dataset	3,095	Seller information
olist_order_reviews_dataset	99,224	Customer review scores
product_category_name_translation	71	Category English translations

3. TECH STACK

Technology	Version	Purpose
Python	3.10	Core programming language
Pandas	2.3.3	Data manipulation and analysis
NumPy	2.2.6	Numerical computations
Matplotlib	3.10.8	Chart generation
Seaborn	0.13.2	Statistical visualizations
SQLite	Built-in	Relational database engine
SQLAlchemy	2.0.48	Database ORM layer
Power BI Desktop	Latest	Interactive dashboard
Git + GitHub	Latest	Version control and portfolio

4. PROJECT WORKFLOW

Step 1 — Environment Setup

Created a Python virtual environment to isolate project dependencies professionally. This is standard practice in every real data team — it prevents library conflicts and makes the project fully reproducible.

```
python -m venv venv venv\Scripts\activate pip install pandas numpy matplotlib seaborn
sqlalchemy openpyxl
```

Step 2 — Data Loading (01_load_data.py)

Loaded all 8 CSV files into a SQLite relational database. This transforms flat files into a proper queryable database with relationships between tables — exactly how real companies store data.

```
conn = sqlite3.connect('ecommerce.db') orders.to_sql('orders', conn,
if_exists='replace', index=False)
```

Step 3 — Exploratory Data Analysis (02_explore_data.py)

Wrote SQL queries across all tables to understand the data before building anything. Found total revenue of R$16M, 91.89% on-time delivery, and November 2017 as peak revenue month.

```
SELECT order_status, COUNT(*) as total, ROUND(COUNT(*)*100.0/ (SELECT COUNT(*) FROM
orders),2) as pct FROM orders GROUP BY order_status ORDER BY total DESC
```

Step 4 — Data Cleaning (03_clean_data.py)

Identified missing values in 3 columns, converted 5 date columns from string to datetime, and created 5 new calculated columns: order_month, order_day_of_week, order_hour, actual_delivery_days, and is_on_time flag. Retained 96,478 delivered orders for analysis.

```
orders['actual_delivery_days'] = ( orders['order_delivered_customer_date'] -
orders['order_purchase_timestamp'] ).dt.days
```

Step 5 — Analysis Charts (04_analysis_charts.py)

Generated 6 professional matplotlib charts: Monthly Revenue Trend (dual-axis), Top 10 Categories, Orders by Day of Week, Review Score Distribution, Delivery Performance Donut, and Orders by Hour of Day. All saved to the outputs folder at 150 DPI.

```
fig, ax1 = plt.subplots(figsize=(14, 5)) ax2 = ax1.twinx()
ax1.fill_between(monthly['order_month'], monthly['revenue'], alpha=0.3)
```

Step 6 — Power BI Export (05_export_for_powerbi.py)

Exported 6 structured Excel files optimized for Power BI: monthly revenue, category performance, delivery by month, review analysis, seller performance, and full dataset with 115,720 rows — all with clean column names and proper data types.

```
full_data.to_excel('outputs/06_full_dataset_powerbi.xlsx', index=False) print(f'Full
Dataset: {len(full_data):,} rows')
```

Step 7 — Power BI Dashboard

Built a 3-page interactive dashboard with cross-filtering and synced slicers. Page 1: Executive Summary. Page 2: Delivery and Seller Analysis. Page 3: Customer Insights. Custom amber color theme applied for unique visual identity.

```
Page 1: KPI Cards + Revenue Trend + Top Categories + Donut + Slicer Page 2: On-Time % +
Delivery Days + Late Orders (RED) + Seller Cities Page 3: Day of Week + Hour Pattern +
Review Trends + Payment Methods
```

Step 8 — Executive Insight Report (07_ai_insights.py)

Generated a professional 5-section executive report by pulling live metrics from the SQLite database and structuring them into business-level narrative with strategic recommendations.

```
Sections: Executive Summary | KPI Highlights | Critical Insights Risk Areas | Strategic
Recommendations
```

5. KEY BUSINESS INSIGHTS DISCOVERED

#	Insight	Finding	Business Impact
1	Revenue Growth	340% growth 2016-2018	Validates expansion strategy
2	Top Category	Health & Beauty R$1.25M	Focus seller acquisition here
3	Delivery Rate	91.9% on-time, 7,826 late	Logistics improvement needed
4	Peak Ordering	Monday 4PM	Prime marketing window
5	Payment Method	73.77% credit card	Critical infrastructure risk
6	Customer Score	4.07/5 avg — bimodal split	Delivery drives dissatisfaction
7	Top Seller City	Sao Paulo dominates	Geographic concentration risk

6. POWER BI DASHBOARD — 3 PAGES

Page 1 — Executive Summary

- Total Revenue: R$15.42M | Total Orders: 96K
- Avg Delivery Days: 13.81 | Avg Review Score: 4.07/5
- Monthly Revenue Trend — line chart showing growth 2016 to 2018
- Top Product Categories by Revenue — sorted horizontal bar chart
- On-Time vs Late Deliveries — donut chart (91.89% vs 8.11%)
- Avg Review Score by Month — column chart showing consistency
- Month Slicer — filters entire page interactively

Page 2 — Delivery & Seller Analysis

- Overall On-Time Delivery %: 89.48% | Total Late Orders: 7,826
- On-Time Delivery % by Month — trend line chart
- Average Delivery Days by Month — showing improvement over time
- Late Orders by Month — red bars highlighting problem months
- Top Cities by Seller Revenue — Sao Paulo clearly dominates

Page 3 — Customer Insights

- Orders by Day of Week — Monday peak at 15,701 orders
- Peak Ordering Hours — 4PM consistently highest across all months
- 5-Star vs 1-Star Reviews by Month — dual line trend
- Orders by Payment Method — credit card at 73.77%

7. SKILLS DEMONSTRATED

Skill	What Was Demonstrated
SQL	Multi-table JOINs, CTEs, aggregations, subqueries, window functions
Python	Data loading, cleaning, transformation, visualization, automation
Pandas	DataFrames, groupby, merge, datetime operations, filtering
Matplotlib	Dual-axis charts, color mapping, professional formatting
SQLite	Database design, table relationships, query optimization
Power BI	Data modeling, DAX basics, cross-filtering, slicers, themes
Git / GitHub	Version control, repository management, documentation
EDA	Exploratory data analysis — pattern discovery before modeling
Data Cleaning	Missing value handling, type conversion, feature engineering
Business Insight	KPI identification, executive reporting, strategic recommendations

8. STRATEGIC RECOMMENDATIONS

Recommendation 1 — Launch Monday Afternoon Campaigns	Schedule all email marketing, push notifications, and promotional offers to go live on Mondays between 14:00 and 16:00. Historical ordering patterns show this window has the highest customer intent, maximizing campaign ROI without additional ad spend.
Recommendation 2 — Implement Delivery SLA Monitoring	Build a real-time delivery performance tracker that flags orders at risk of being late before SLA breach. Proactive customer communication for at-risk orders reduces 1-star reviews by up to 30% even when delivery is late, as customers respond positively to transparency.
Recommendation 3 — Double Down on Top 3 Revenue Categories	Allocate 60% of seller acquisition budget to Health and Beauty, Watches and Gifts, and Bed Bath Table. These categories already have proven customer demand and trust. More sellers means better pricing, more choice, and higher conversion without new customer acquisition cost.

9. PROJECT FILE STRUCTURE

```
project1_ecommerce/ ■ ■■■ 01_load_data.py # Load CSVs into SQLite database ■■■
02_explore_data.py # SQL-based exploratory analysis ■■■ 03_clean_data.py # Data
cleaning and enrichment ■■■ 04_analysis_charts.py # Generate 6 analysis charts ■■■
05_export_for_powerbi.py # Export clean Excel files for Power BI ■■■
06_fix_dayofweek.py # Day-of-week sort order fix ■■■ 07_ai_insights.py # Auto-generate
executive report ■ ■■■ outputs/ ■ ■■■ 01_monthly_revenue_trend.png ■ ■■■
02_top_categories_revenue.png ■ ■■■ 03_orders_by_day.png ■ ■■■
04_review_distribution.png ■ ■■■ 05_delivery_performance.png ■ ■■■
06_orders_by_hour.png ■ ■■■ 07_ai_executive_report.txt ■ ■■■ 01_monthly_revenue.xlsx ■
■■■ 02_category_performance.xlsx ■ ■■■ 03_delivery_performance.xlsx ■ ■■■
04_review_analysis.xlsx ■ ■■■ 05_seller_performance.xlsx ■ ■■■
06_full_dataset_powerbi.xlsx ■ ■■■ E-Commerce Performance Dashboard.pbix ■■■
requirements.txt ■■■ PROJECT_DOCUMENTATION.md ■■■ README.md
```

10. PROJECT LINKS

Resource	Link
GitHub Repository	https://github.com/Sabarivenkatesh3/ecommerce-analytics-dashboard
Dataset Source	https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce
LinkedIn Profile	https://www.linkedin.com/in/sabarivenkatesh-kathirvel

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Sabarivenkatesh Kathirvel | Aspiring Data Analyst | SQL · Python · Power BI · Data Cleaning · EDA · Business Intelligence